

DAY 10



Restaurant Database SQL Project

Project Overview

This project is based on a sample database called Restaurant_db. It contains a table menu_items that stores details of different dishes served in the restaurant. Using SQL queries, we perform data exploration and analysis to answer common business questions.

The goal is to demonstrate how SQL can be used for:

Retrieving data ,Filtering records ,Sorting results,Aggregating (sum, avg, min, max)

Column Name	Description
Item id	Unique ID for each menu item
Item name	Name of the dish
Category	Category (e.g., Italian, Chinese, Dessert, Beverages)
Price	Price of the dish

use restaurant_db; DATABASE THAT WE USED HERE

1. View all items

```
select * from menu_items;
```

2. Count total menu items

```
select count(*) from menu_items;
```

3. Find maximum and minimum priced items

```
select  
max(price) as Max_price,  
min(price) as Min_price  
from menu_items;
```

4. Count Italian items

```
select count(category) as Italian_item  
from menu_items  
where category='Italian';
```

5. List Italian items sorted by price

```
select * from menu_items
where category ='Italian'
order by price;
```

```
select * from menu_items
where category ='Italian'
order by price desc;
```

6.how many disheds in each ctegrory

```
select * from menu_items;
select category,count(item_name) as No_of_dishes
from menu_items
group by category;
```

7.average dishes price in each category

```
select category,avg(price) as No_of_dishes
from menu_items
group by category
```

```
select * from order_details;      Another Database that we used here
```

1.what is the data range of the table

```
select Max(order_date), min(order_date) from order_details;
```

2.how many orders were made within this range

```
select count(distinct order_id) from order_details;
```

3.how many items were order with in this range

```
select count(*) from order_details;
select
order_id,
count(item_id) as Num_items
from order_details
group by order_id
order by Num_items desc;
```

4.how many orders has more than 12 items

```
select count(*) from
(select
order_id,
count(item_id) as Num_items
from order_details
group by order_id
having Num_items > 12) as Num_orders
```

```
select * from menu_items;  
select * from order_details;
```

```
select * from order_details od left join menu_items mi  
on od.item_id= mi.menu_item_id;
```

5.I want to keep the all the transactions so that i have to mek the join as the left join

```
select item_name, category,count(order_details_id) as num_purchases  
from order_details od left join menu_items mi  
on od.item_id= mi.menu_item_id  
group by item_name , category  
order by num_purchases;
```

6.Top 5 order that spent the most money

```
select order_id,  
sum(price) as Total_spend  
from order_details od left join menu_items mi  
on od.item_id= mi.menu_item_id  
group by order_id  
order by Total_spend desc  
limit 5
```